

$$= 0.729 \text{ T}$$

- 4a. (Reasoning: the magnetic force must be downward, to balance the weight of the mass on the other end. Use Fleming's left-hand rule.)

B to A

- b. Take moments about pivot
 Anticlockwise moment = clockwise moment
 $BIL \times 8.0 \times 10^{-2} = mg \times 7.0 \times 10^{-2}$

$$\begin{aligned}
 B &= \frac{mg \times 7.0 \times 10^{-2}}{IL \times 8.0 \times 10^{-2}} \\
 &= \frac{2.0 \times 10^{-3} \times 9.81 \times 7.0 \times 10^{-2}}{3.0 \times 6.0 \times 10^{-2} \times 8.0 \times 10^{-2}} \\
 &= 9.5 \times 10^{-2} \text{ T}
 \end{aligned}$$

- c.
- A non-horizontal field will generate a magnetic force with a horizontal component.
 - This horizontal component force will not generate any moment about the pivot.
 - So you can't tell what is its magnitude (and therefore you won't know what is the total value of the magnetic field).